

ESE 541 Week 5 & 6 Practice Problems

Problem 1. Define the rectangular function as $R(x) = 1$ for $x \in [-2, 2]$, 0 otherwise:

1. Plot $R(x)$.
2. Plot $R(x + 1)$.

Problem 2. Define the sigmoid function as $S(x) = \frac{1}{1+e^{-x}}$:

1. Show that $S(x)$ can also be written as $\frac{e^x}{e^x+1}$.
2. Show that $S(-x)$ can be written as $\frac{1}{1+e^x}$ and $\frac{e^{-x}}{e^{-x}+1}$.
3. Find an expression for $1 - S(x)$. What is the relationship between $1 - S(x)$ and $S(-x)$?

Problem 3. Let $P(Y = +1) = P(Y = -1) = 0.5$. We are also given:

$$P(X = x|Y = +1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-3)^2}{2}} \quad P(X = x|Y = -1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x+3)^2}{2}}$$

These conditional probabilities can also be expressed as $P(x|+1) \sim \mathcal{N}(3, 1)$ and $P(x|-1) \sim \mathcal{N}(-3, 1)$, where $\mathcal{N}(\mu, \sigma^2)$ represents a normal distribution centered at mean μ with standard deviation σ .

1. Find an expression for $P(X = x, Y = y)$
2. Plot the conditional distributions $P(X = x|Y = +1)$ and $P(X = x|Y = -1)$. On the graph, show the Bayes optimal classification rule.
3. Calculate the mass of $P(X = x|Y = -1)$ that lies on the wrong side of the decision rule

Problem 4. Explain what the following represent in the context of detecting spam email

1. False Positive Rate (FPR)
2. False Negative Rate (FNR)
3. True Positive Rate (TPR)

Problem 5. Suppose we have a dataset of 100 students for which we have access to two features: X_1 ='hours studied', X_2 ='GPA', and Y = 'receive an A'. We fit a logistic regression and produce the following estimated coefficients: $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, and $\hat{\beta}_2 = 1$.

1. Estimate the probability that a student who studies for 40 hours and has a GPA of 3.5 gets an A in the class.
2. How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

Problem 6. Prove that

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

is equivalent to

$$p(X) = \frac{\exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}.$$

Problem 7: Consider a communication system transmitting a variable Y that can take two possible values, either 0 or 1. The communication channel is noisy, so that the receptor receives a noisy version of Y equal to $X = 2Y + \varepsilon$, where ε is a gaussian $\mathcal{N}(-1, \frac{1}{4})$. We also know that the communication system transmits, in expectation, 50% of 0's and 50% of 1's.

1. Write down expressions for $f_0(x) = \Pr(X = x|Y = 0)$ and $f_1(x) = \Pr(X = x|Y = 1)$? What are the values of the priors, $\pi_0 = \Pr(Y = 0)$ and $\pi_1 = \Pr(Y = 1)$?
2. Use Bayes' rule to find the conditional class probabilities $\Pr(Y = 0|X = x)$ and $\Pr(Y = 1|X = x)$.
3. Compute the false positive rate and true positive rate of the classifier.
4. What is the total error rate of the classifier?